BANK MARKETING DATASET ANALYSIS

INTRODUCTION

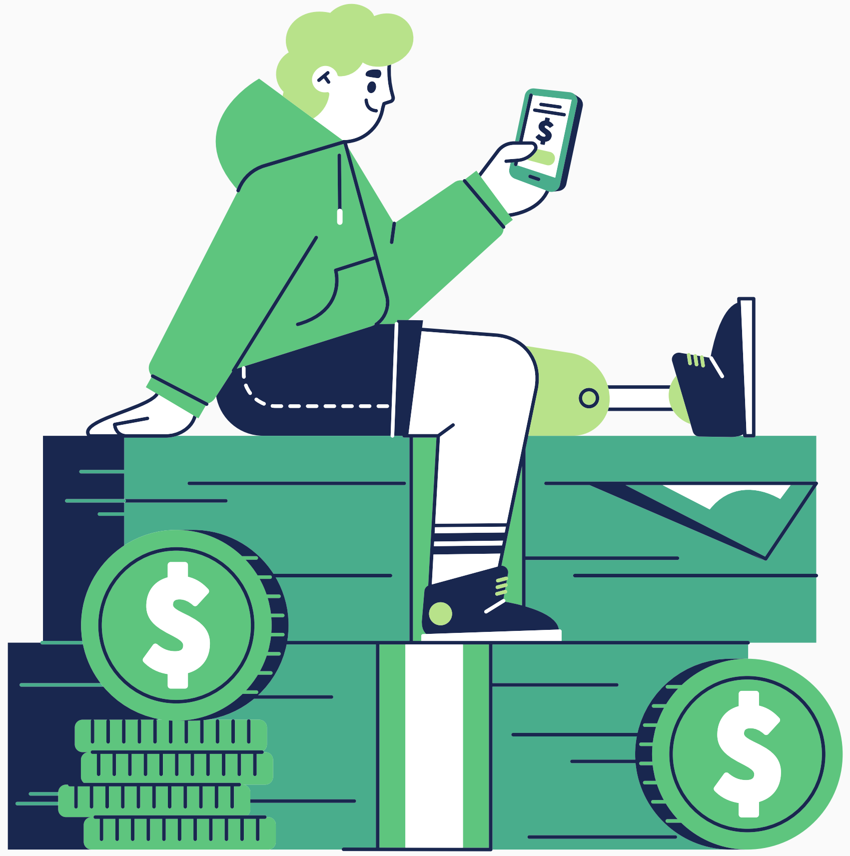
The Bank Marketing dataset contains customer and campaign-related information collected during telephonic marketing campaigns conducted by a Portuguese bank. The objective of this project is to analyze the data and predict whether a customer will subscribe to a term deposit, helping the bank improve marketing efficiency and reduce unnecessary calls.

BUSINESS PROBLEM STATEMENT

The main objective of this project is to predict whether a customer will subscribe to a term deposit based on their demographic, financial, and past marketing information. By identifying high-potential customers in advance, the bank can reduce unnecessary marketing calls, lower operational costs, and focus its efforts on customers who are more likely to respond positively.

VARIABLE DESCRIPTION

The dataset contains both categorical and numerical variables describing customer information and marketing campaign details.
Categorical variables include job, marital status, education, default, housing, loan, contact type, month, day of contact, previous campaign outcome, pdays , current campaign outcome and the target variable (y).
Numerical variables include age, balance, campaign, which represent customer demographics and interaction history.
The target variable y indicates whether the customer subscribed to a term deposit (Yes/No).



DATA OVERVIEW

> The dataset contains 45,211 records and 17 variables.
> There are 3 numerical and 14 categorical features.
> All columns have non-null values, indicating no missing data initially.
> Memory usage of the dataset is approximately 5.9 MB.
> The target variable y represents customer subscription (Yes/No).

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 45211 entries, 0 to 45210
Data columns (total 17 columns):
 #   Column     Non-Null Count  Dtype 
---  ------     --------------  ----- 
 0   age        45211 non-null  int64 
 1   job        45211 non-null  object
 2   marital    45211 non-null  object
 3   education  45211 non-null  object
 4   default    45211 non-null  object
 5   balance    45211 non-null  int64 
 6   housing    45211 non-null  object
 7   loan       45211 non-null  object
 8   contact    45211 non-null  object
 9   day        45211 non-null  int64 
 10  month      45211 non-null  object
 11  duration   45211 non-null  int64 
 12  campaign   45211 non-null  int64 
 13  pdays      45211 non-null  int64 
 14  previous   45211 non-null  int64 
 15  poutcome   45211 non-null  object
 16  y          45211 non-null  object
dtypes: int64(7), object(10)
memory usage: 5.9+ MB
```

DATA CLEANING

KEY STEPS PERFORMED

- STANDARDIZED COLUMN NAMES FOR CONSISTENCY
- IDENTIFIED HIDDEN MISSING VALUES REPRESENTED AS "UNKNOWN"
- CONVERTED "UNKNOWN" VALUES TO NAN FOR UNIFORM MISSING VALUE HANDLING
- IMPUTED MISSING VALUES USING FORWARD FILL AND BACKWARD FILL TECHNIQUES
- VERIFIED THAT NO MISSING VALUES REMAINED AFTER CLEANING
- HANDLED SPECIAL VALUES SUCH AS PDAYS = 999 TO AVOID MISLEADING STATISTICAL INTERPRETATION

DATA CLEANING

NULL VALUES

```
age          0
job          0
marital      0
education    0
default      0
balance      0
housing      0
loan         0
contact      0
day          0
month        0
duration     0
campaign     0
pdays        0
previous     0
poutcome     0
y            0
dtype: int64
```

UNKNOWN VALUES

	0
age	0
job	288
marital	0
education	1857
default	0
balance	0
housing	0
loan	0
contact	13020
day	0
month	0
duration	0
campaign	0
pdays	0
previous	0
poutcome	36959
y	0

UNIQUES VALUES

	Column_Name	Unique_Values	Unique_Count
0	job	[management, technician, entrepreneur, blue-co...	12
1	marital	[married, single, divorced]	3
2	education	[tertiary, secondary, unknown, primary]	4
3	default	[no, yes]	2
4	housing	[yes, no]	2
5	loan	[no, yes]	2
6	contact	[unknown, cellular, telephone]	3
7	month	[may, jun, jul, aug, oct, nov, dec, jan, feb, ...	12
8	poutcome	[unknown, failure, other, success]	4
9	y	[no, yes]	2

DATA CLEANING

NAN VALUE TREATMENT

	age	job	marital	education	default	balance	housing	loan	contact	day	month	duration	campaign	pdays	previous	poutcome	y
0	58	management	married	tertiary	no	2143	yes	no	cellular	5	may	261	1	-1	0	failure	no
1	44	technician	single	secondary	no	29	yes	no	cellular	5	may	151	1	-1	0	failure	no
2	33	entrepreneur	married	secondary	no	2	yes	yes	cellular	5	may	76	1	-1	0	failure	no
3	47	blue-collar	married	secondary	no	1506	yes	no	cellular	5	may	92	1	-1	0	failure	no
4	33	management	single	secondary	no	1	no	no	cellular	5	may	198	1	-1	0	failure	no

NAN VALUES

```
age          0
job          0
marital      0
education    0
default      0
balance      0
housing      0
loan         0
contact      0
day          0
month        0
duration     0
campaign     0
pdays        0
previous     0
poutcome     0
y            0
dtype: int64
```

CHECKING THE NAN VALUES

PREPROCESSING

Removal of duration (Data Leakage Prevention)

- Duration indicates the length of the marketing call
- This information is available only after the call ends
- Including it would cause data leakage
- Data leakage leads to unrealistic and inflated model performance
- Therefore, the duration column was removed from the dataset

EXPLORATORY DATA ANALYSIS (EDA)

CONTENT

- Performed exploratory analysis to understand:
- Target variable distribution
- Numerical feature behavior
- Presence of outliers
- Class imbalance
- EDA insights guided preprocessing and modeling decisions

Outcome: Data behavior understood before model building

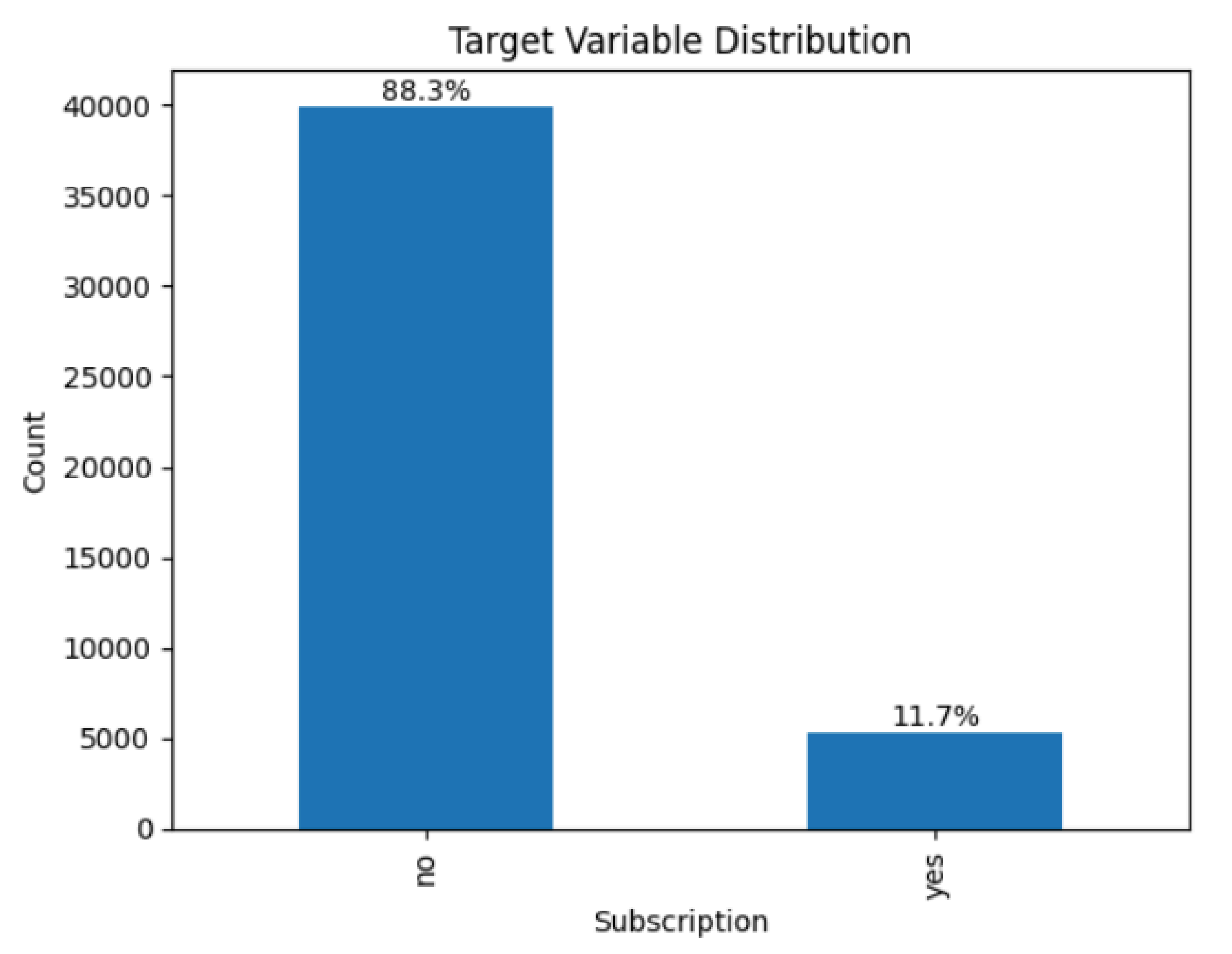
EXPLORATORY DATA ANALYSIS- TARGET VARIABLE

PURPOSE

- To analyze customer subscription behavior
- To check class balance

Key Observation

- Target variable y is highly imbalanced
- Majority customers did not subscribe
- Small percentage subscribed to term deposit



Outcome: Identified class imbalance in target variable

EDA - UNIVARIATE ANALYSIS (NUMERICAL FEATURES)

To understand distribution, skewness, and spread of numerical variables

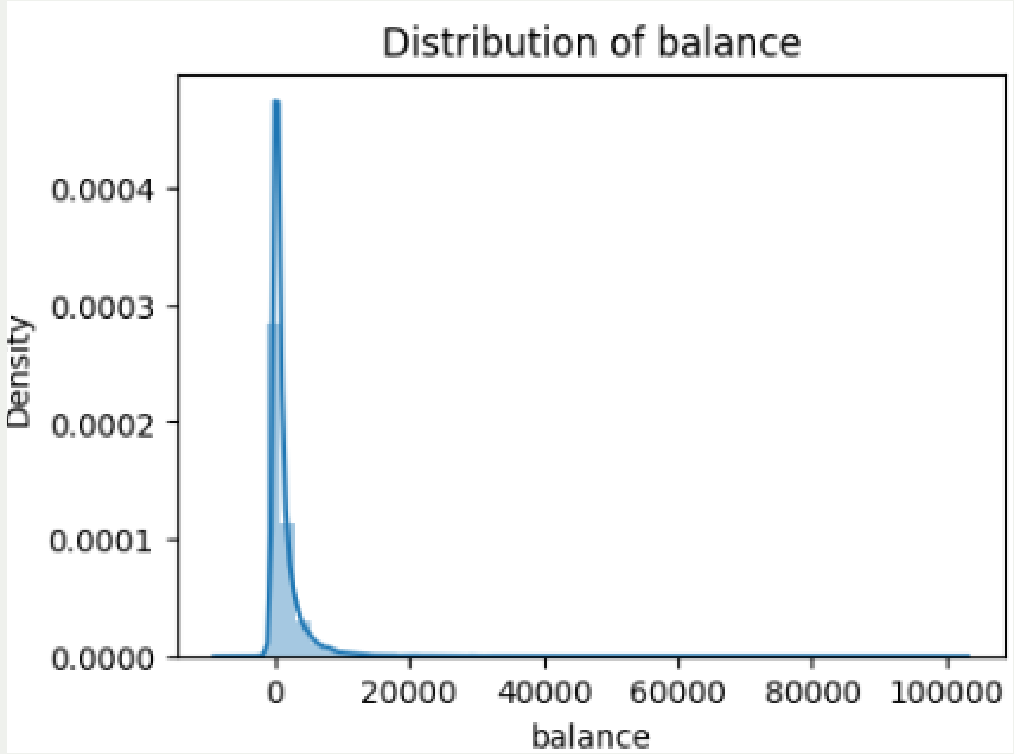
CONTENT

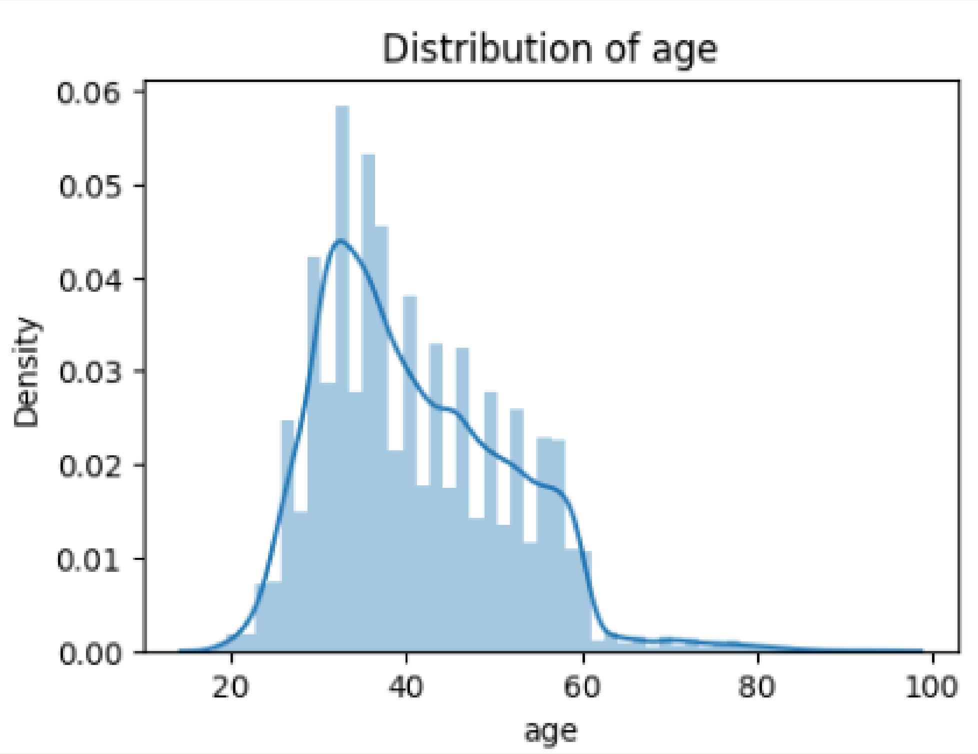
- Distribution plots created for numerical features:
  - age
  - balance
  - day
  - campaign
  - pdays
  - previous
- Used histograms / density plots to observe data patterns

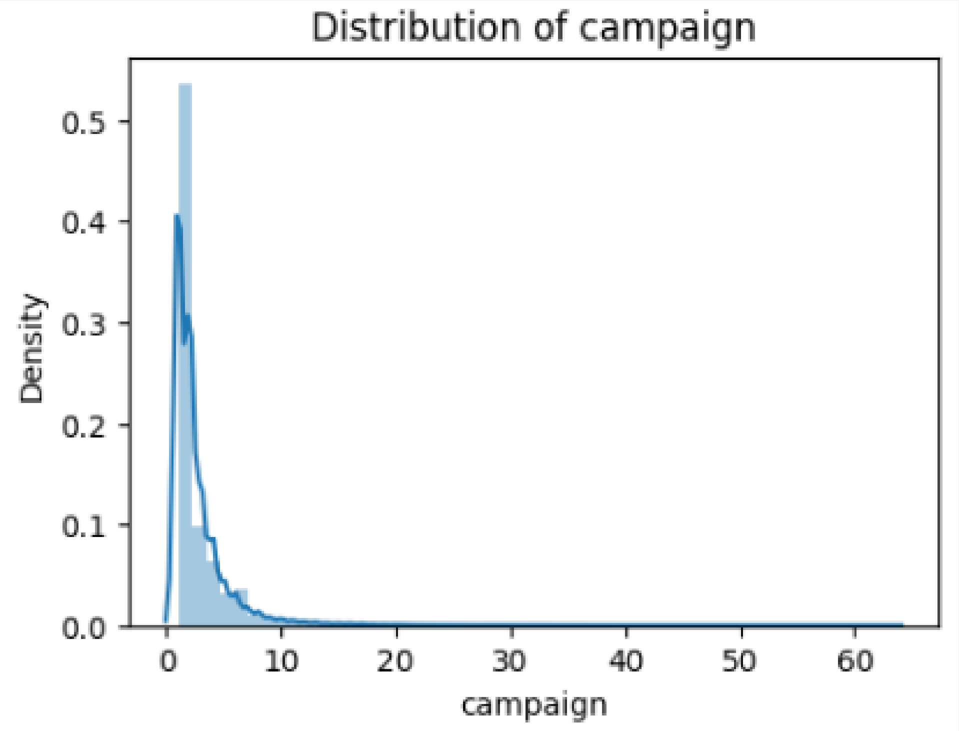
Insight

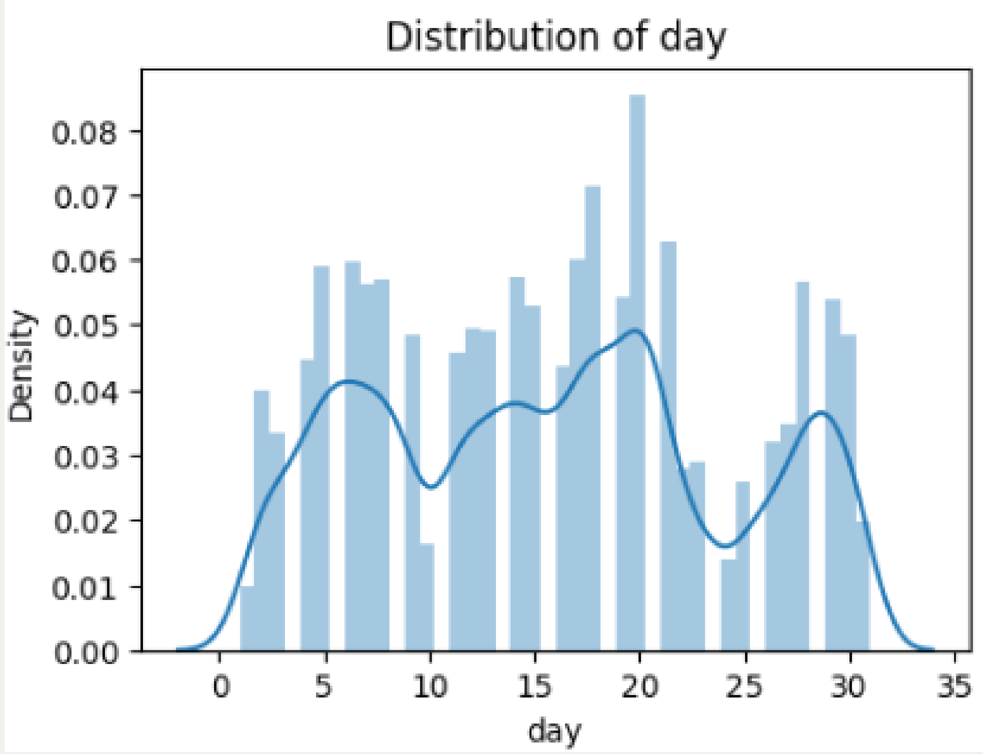
- Several variables show right-skewed distributions
- Indicates need for scaling and outlier handling

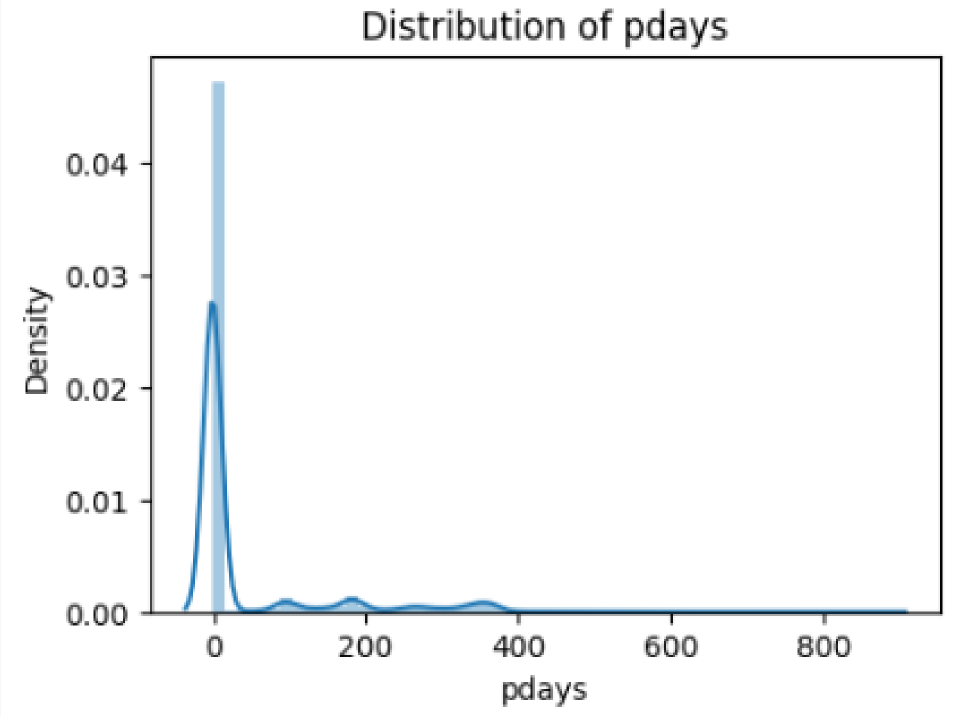
EDA - UNIVARIATE ANALYSIS (NUMERICAL FEATURES)

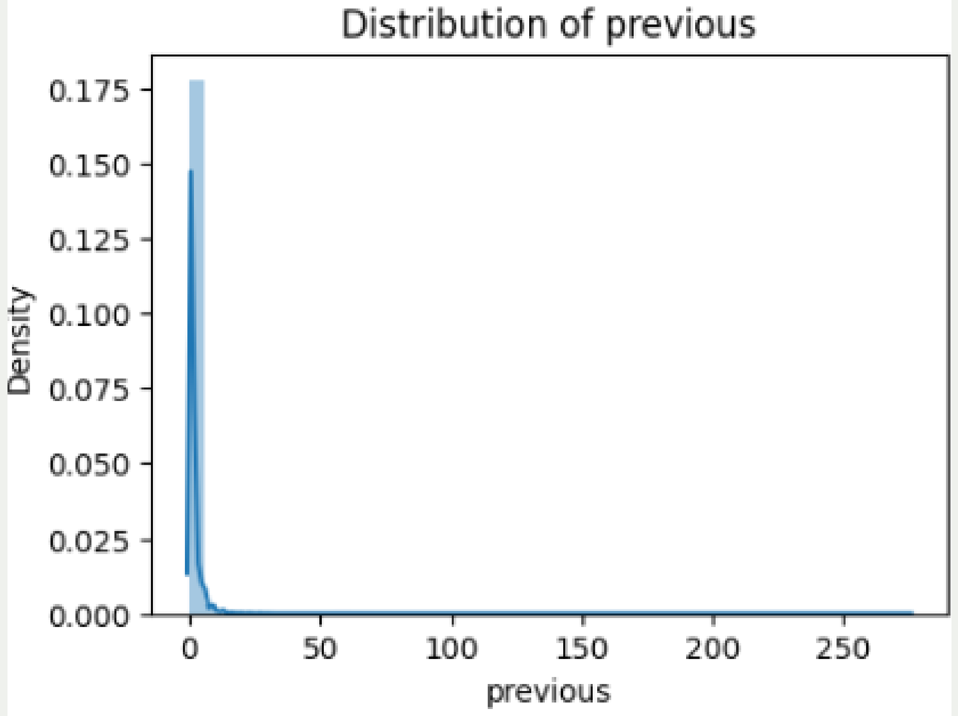












OUTCOME: IDENTIFIED SKEWNESS AND NON-NORMAL DISTRIBUTIONS

OUTLIER DETECTION

Purpose

To detect extreme values and data spread Explanation

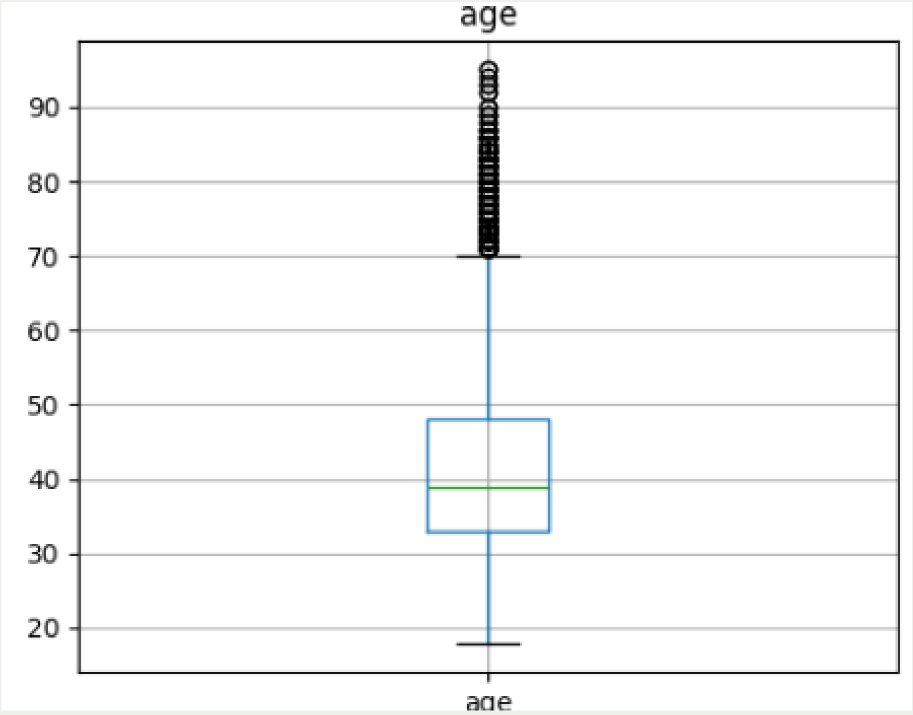
- Boxplot components:
  - Median
  - IQR
  - Outliers

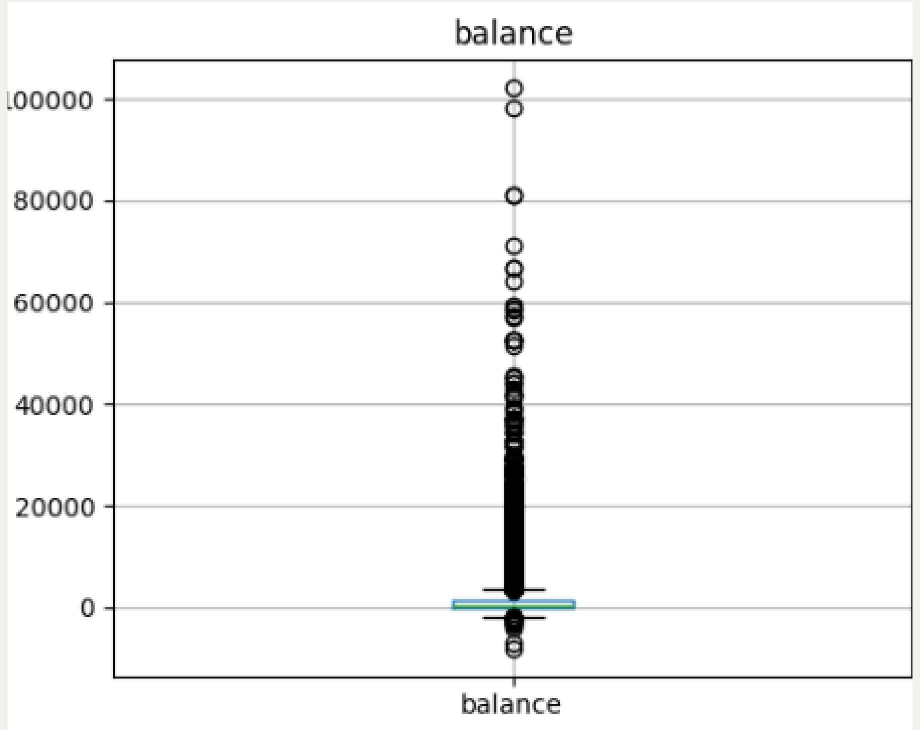
Key Findings

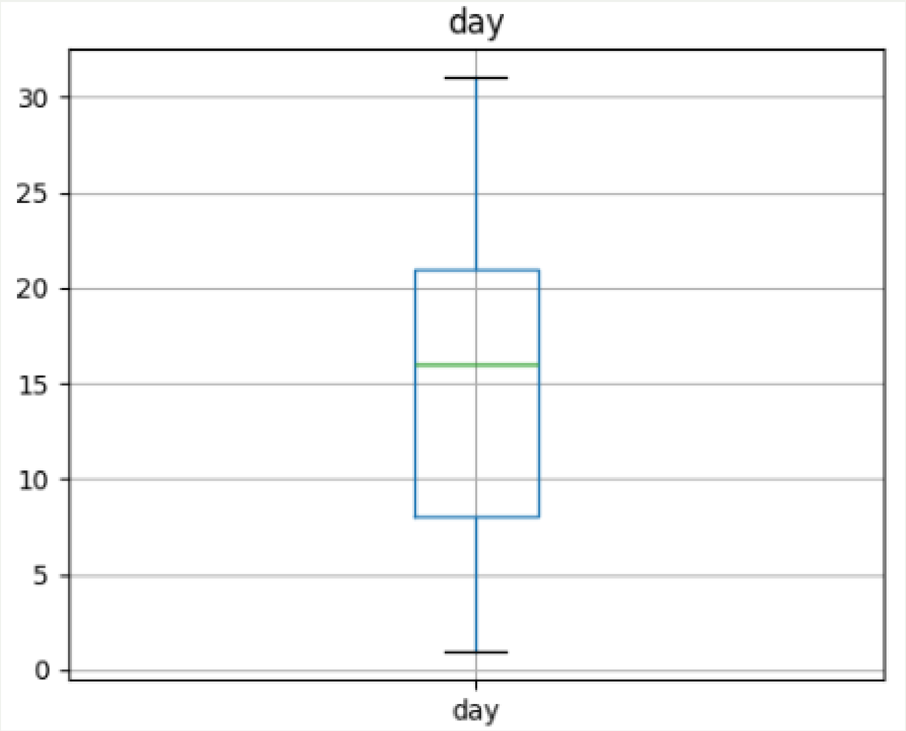
- Significant outliers in:
  - balance
  - campaign
  - pdays
  - previous
- age contains valid but extreme values

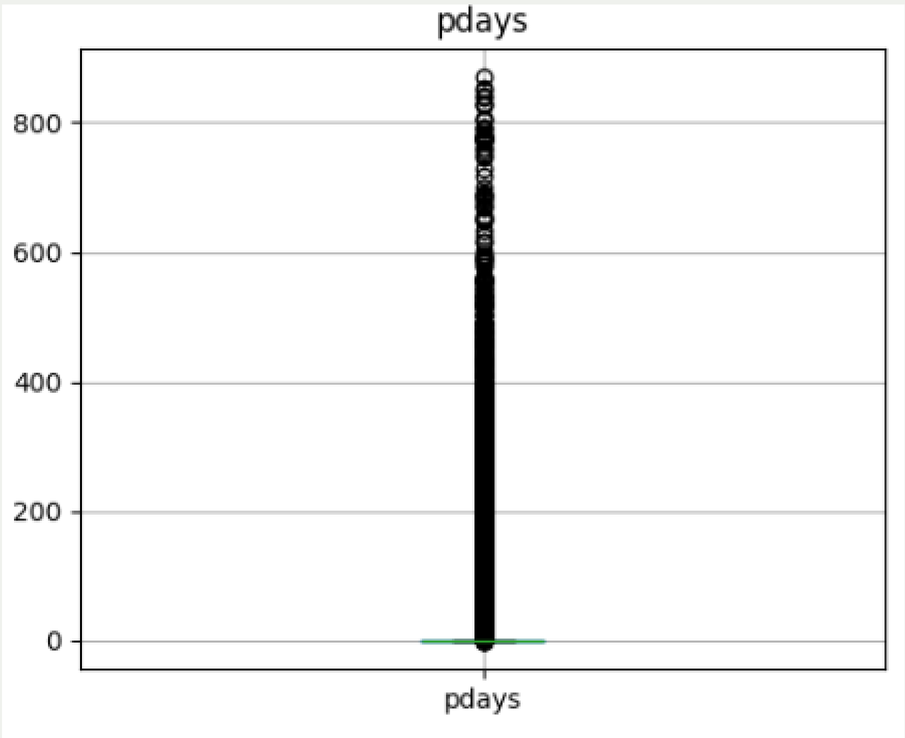
```
age: 487 outliers
balance: 4729 outliers
day: 0 outliers
campaign: 3064 outliers
pdays: 8257 outliers
previous: 8257 outliers
```

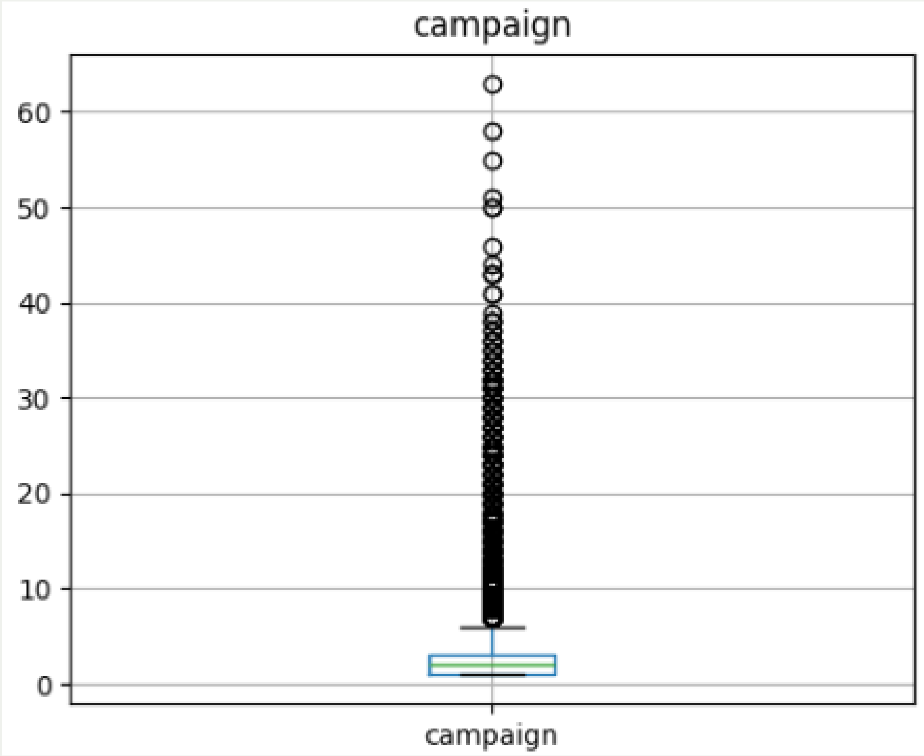
OUTLIER DETECTION - BOXPLOTS

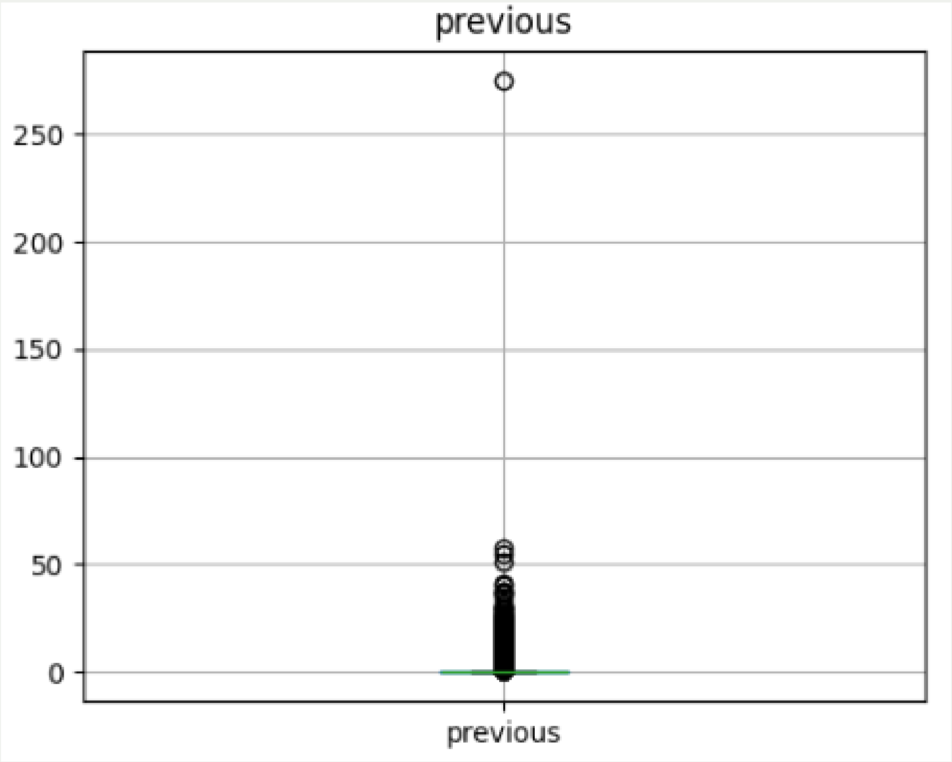












OUTCOME: VISUAL CONFIRMATION OF EXTREME VALUES

OUTLIER TREATMENT

Techniques Applied

- Age capped between 18–90
- Campaign capped at 95th percentile
- Previous capped at 95th percentile
- Created binary feature pdays_contacted

Why Treatment Was Needed

- Reduce influence of extreme values
- Improve model stability

	proportion
y	
no	0.883015
yes	0.116985

dtype: float64

Outcome: Reduced outlier impact without data loss

CLASS IMBALANCE CONFIRMATION

Purpose

To reconfirm imbalance after preprocessing

Observation

- Target variable remains highly imbalanced
- Accuracy alone is insufficient for evaluation

	proportion
y	
no	88.3
yes	11.7

dtype: float64

Outcome: Guided metric selection (Precision, Recall, F1)

FEATURE ENCODING & SCALING

Feature Encoding

- Target variable (y) ko features se separate kiya
- Numerical aur categorical columns identify kiye
- Categorical features ko One-Hot Encoding se convert kiya
- drop_first = True use kiya to avoid multicollinearity

```
X_train.shape, X_test.shape
```

```
((36168, 38), (9043, 38))
```

Train-Test Split

- Data ko 80% training aur 20% testing me divide kiya
- Stratified split use kiya to maintain class imbalance

Feature Scaling

- Numerical features par Min–Max Scaling apply ki
- Values ko 0–1 range me normalize kiya
- Scaling train–test split ke baad ki gayi
- Data leakage avoid kiya

MACHINE LEARNING ALGORITHMS

Algorithms Implemented

- Decision Tree Classifier
- Random Forest Classifier
- Linear Support Vector Machine (SVM)

Problem Type

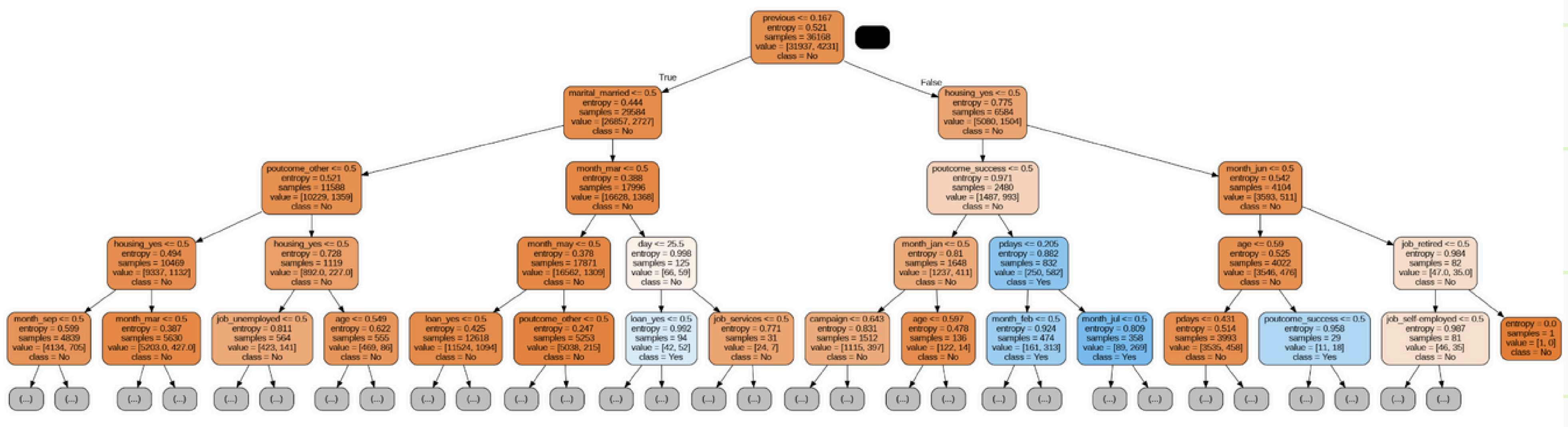
- Binary Classification
- Predicting customer subscription (Yes / No)

Outcome: Multiple models trained for performance comparison

DECISION TREE CLASSIFIER

Why Decision Tree?

- Simple and interpretable model
- Handles non-linear relationships
- Good baseline classifier



Model Configuration

- Criterion: Entropy
- Feature selection: sqrt
- Tree depth limited for interpretability

Outcome: Interpretable baseline model built

DECISION TREE - MODEL EVALUATION

Performance Metrics

- Accuracy: 83.6%
- High performance for No class
- Lower recall for Yes class due to imbalance

Key Observation

Model struggles to correctly predict minority class

Outcome: Baseline model highlights class imbalance issue

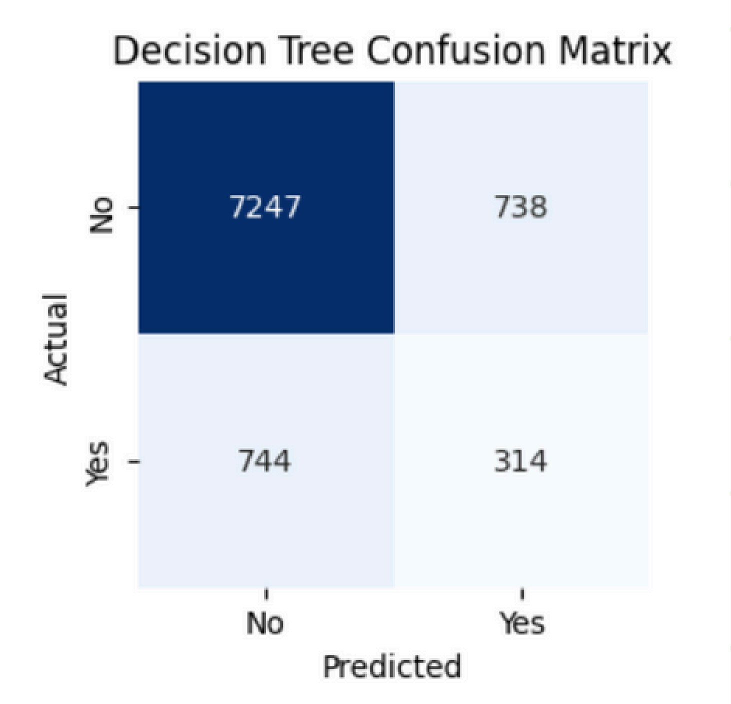
```
Accuracy: 0.8361163330753069

Confusion Matrix:
 [[7247  738]
 [ 744  314]]

Classification Report:
              precision    recall  f1-score   support

          no       0.91      0.91      0.91      7985
         yes       0.30      0.30      0.30      1058

    accuracy                           0.84      9043
   macro avg       0.60      0.60      0.60      9043
weighted avg       0.84      0.84      0.84      9043
```



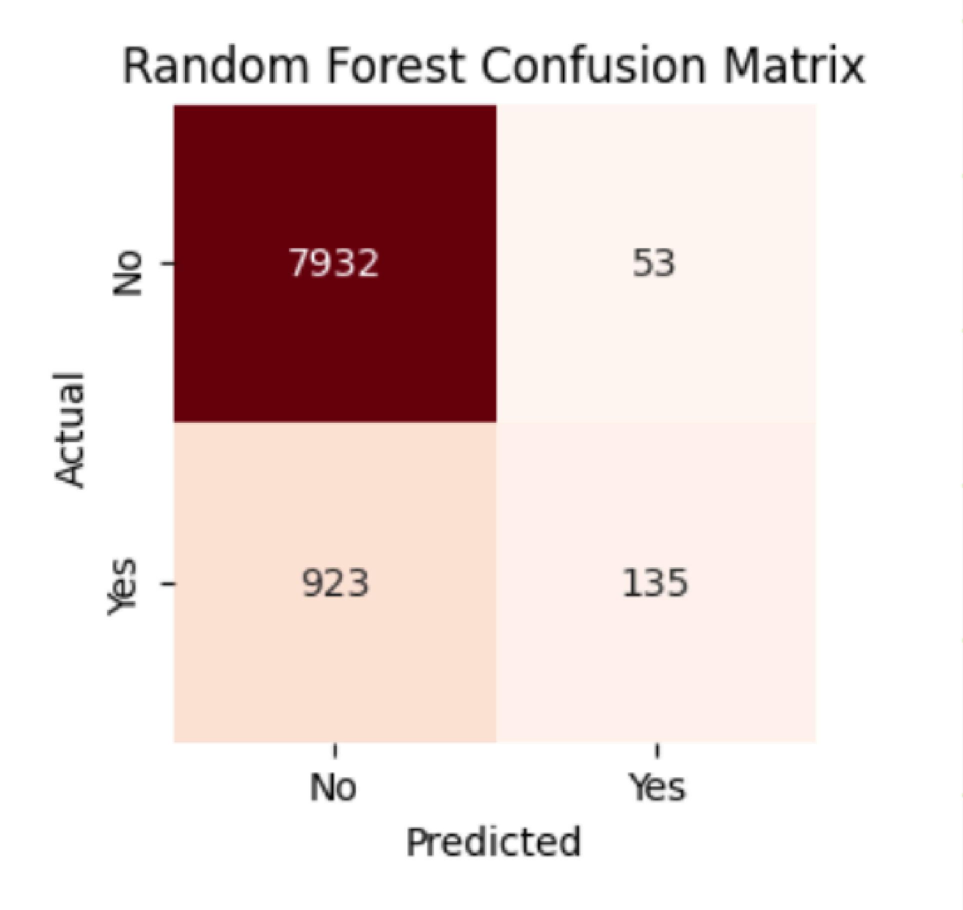
RANDOM FOREST CLASSIFIER

Why Random Forest?

- Ensemble of multiple decision trees
- Reduces overfitting
- Improves generalization

Model Highlights

- 200 trees used
- Depth controlled to avoid overfitting
- Out-of-Bag (OOB) evaluation applied



Outcome: Best performing model overall

RANDOM FOREST - INSIGHTS

Performance Summary

- Accuracy: 89.2%
- Very strong prediction for No class
- Still limited recall for Yes class

Insight

- Class imbalance affects minority predictions
- Advanced techniques like SMOTE can improve results

Ensemble model improves robustness

SUPPORT VECTOR MACHINE (LINEAR)

Why Linear SVM?

- Dataset is large and high-dimensional
- Efficient after one-hot encoding
- Lower risk of overfitting

Outcome: Linear baseline model evaluated

SVC

```
SVC(kernel='linear', max_iter=2000)
```

```
Test Accuracy: 0.572265840981975

Confusion Matrix:
 [[4819 3166]
 [ 702  356]]

Classification Report:
```

	precision	recall	f1-score	support
no	0.87	0.60	0.71	7985
yes	0.10	0.34	0.16	1058
accuracy			0.57	9043
macro avg	0.49	0.47	0.43	9043
weighted avg	0.78	0.57	0.65	9043

MODEL COMPARISON & CONCLUSION

Final Conclusion

- Random Forest performed best overall
- Class imbalance remains a challenge
- Future work: SMOTE / class weighting

MODEL	ACCURACY
Decision Tree	83.6%
Random Forest	89.2%
Linear SVM	57.2%

Outcome: Random Forest selected as the optimal model

RESULTS & DISCUSSION

Overall Findings

- Multiple machine learning models were evaluated
- Performance comparison was done using accuracy, precision, recall, and F1-score

Model-wise Insights - Decision Tree

- Accurately identifies non-subscribers (No class)
- Limited ability to detect actual subscribers
- Useful for reducing unnecessary marketing calls

RESULTS & DISCUSSION

Model-wise Insights - Random Forest

- Best overall performance among all models
- Higher precision for potential subscribers (Yes class)
- More reliable and stable predictions

Model-wise Insights - Linear SVM

- Lowest accuracy among evaluated models
- Performance affected by class imbalance
- Less suitable for deployment

Outcome: Random Forest selected for deployment due to best performance

This analysis enables the bank to focus resources on high-probability customers.

THANK YOU

BY -

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